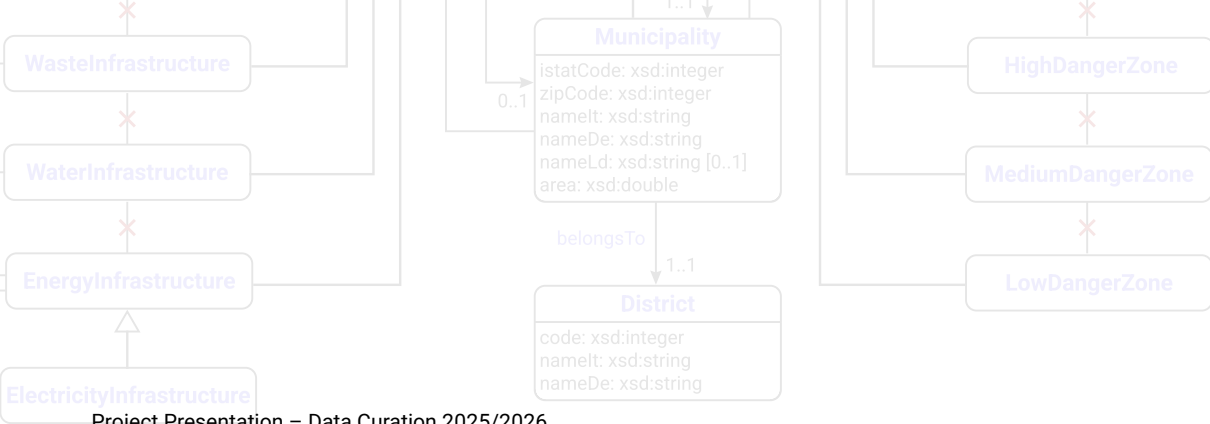




Project Presentation – Data Curation 2025/2026

Semantic Integration of Geospatial Data for Infrastructure Vulnerability Assessment

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September 2026

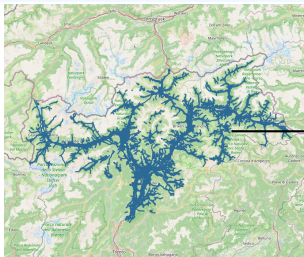
Introduction

- ▶ Core Goals:

- Analyze exposure of critical infrastructure to natural hazards within the Bolzano region.
- Identify infrastructure assets located within high-risk landslide or avalanche zones.
- Data curation pipeline from data profiling to exposing the VGK through an application.

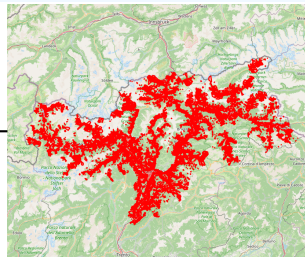
- ▶ Example Application Questions:

- Which water nodes or electricity lines sit in high-risk hazard zones?
- Which municipalities face the highest concentration of exposed critical assets?



Infrastructure Assets

Integrate



Hazard Zones

Selected Data Sources

- ▶ **Infrastructure:**

- `bz_infrastructure_line` will be used as the source for lines-based infrastructure.
- `bz_infrastructure_node` provides information about point-like infrastructure.

- ▶ **Hazards:**

- `bz_hazard_landslide` includes information about landslide-prone areas.
- `bz_hazard_avalanche` covers areas at risk of avalanches.

- ▶ **Administrative:**

- `bz_admin_municipality` to provide spatial context and assign risk to specific municipalities.

Data Profiling – Elementary Data Analysis

- ▶ Data Profiling Stack:

- Metanome Platform for type inference and simple statistics.
- YData Profiling for exploratory analysis, type distributions, and correlation matrices.

- ▶ Integrated Datasets:

- bz_infrastructure_line (47,551 rows, 11 types, MULTILINESTRING).
- bz_infrastructure_node (4,223 rows, 10 types, POINT)
- bz_hazard_landslide (23,737 rows, 4 danger levels, 5 process types, POLYGON)
- bz_hazard_avalanche (13,071 rows, 4 danger levels, 5 process types, POLYGON)
- bz_admin_municipality (116 rows, municipal boundaries and district data, POLYGON)

- ▶ Initial Data Observations:

- Geometries stored in Well-Known Text format.
- Presence of multiple uninformative columns, either missing or constant.
- Float-formatted integer identifiers in all tables.

Data Profiling – Dependency Discovery

- ▶ Data Profiling Stack:

- Metanome Platform used for all dependency discover algorithms.

- ▶ Unique Column Combinations: Using HyUCC algorithm.

- Discovered only single-column candidate keys across tables: `ogc_fid`, `gml_id`, `objectid`.
 - Selected `objectid` as primary key due to global uniqueness across tables.

- ▶ Functional Dependencies: Using HyFD algorithm.

- `code` → {`bez_d`, `bez_i`} Infrastructure type names redundant in asset tables.
 - `code` → {`gefahr`, `pericolo`} Hazard classifications redundant.
 - `distr_code` → {`distr_de`, `distr_it`} District names redundant in municipal table.

- ▶ Inclusion Dependencies: Using BINDER algorithm:

- Confirmed foreign keys `istat_code` are strictly included in `Municipality.istat_code`.

Relational Database Schema Design

- ▶ Normalization and Entity Extraction:

- Extracted 4 relations to avoid duplication:

InfrastructureType, DangerLevel, HazardProcess, District

- Maintained 5 core spatial relations:

InfrastructureLine, InfrastructureNode, LandslideHazard, AvalancheHazard, Municipality

- ▶ Schema Constraints and Data Types:

- Primary keys enforced on `code` and `istat_code`.

- Foreign keys enforced for municipality links and normalized tables.

- Transformed trailing float identifiers e.g., 21001.0, to standard `INTEGER`.

- Geometries mapped to PostGIS `GEOMETRY` type.

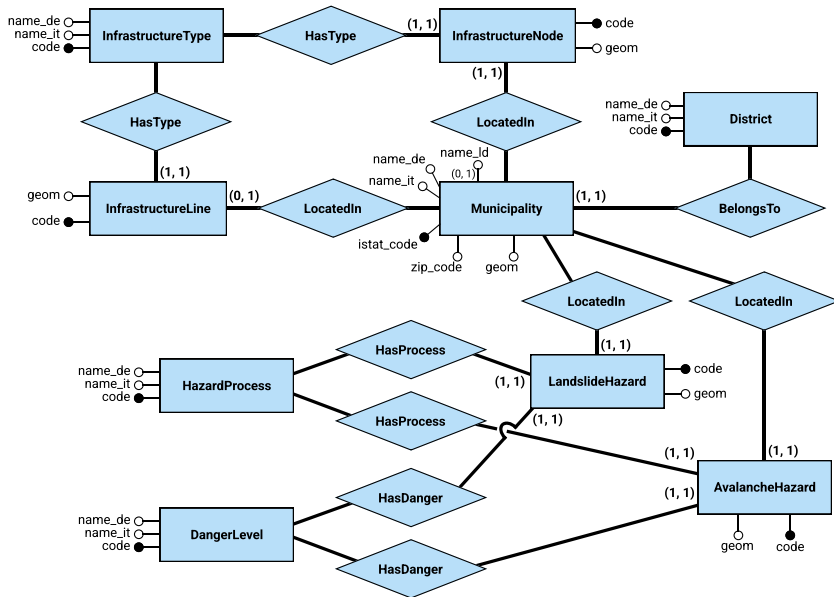
- ▶ Attribute Cleanup:

- Dropped empty and constant fields: `blp`, `codealt`, `status_gzp`, `id_phen`.

- Dropped UI-only and survey fields: `nda_text_*`, `nda_link_*`, `x_label`, `y_label`, `bearbeitungstiefe`.

- Omitted derived metrics, `shape_len` and `area`, to calculate them dynamically.

Relational Database Schema Design – Cont.



Domain Ontology Modeling

- ▶ **Ontology Architecture:**

- 18 classes, 5 object properties, and 20 data properties.
- `GeometricObject` → (disjoint) `Infrastructure`, `HazardZone`, `Municipality`.

- ▶ **Specialized Subclasses:**

- **Structural** (`Line`, `Node`) and **Thematic** (`Energy`, `Water`, `Waste`, `Communication`, `Electricity`).
- **Type** (`LandslideZone`, `AvalancheZone`) and **Severity** (`VeryHigh`, `High`, `Medium`, `Low`)

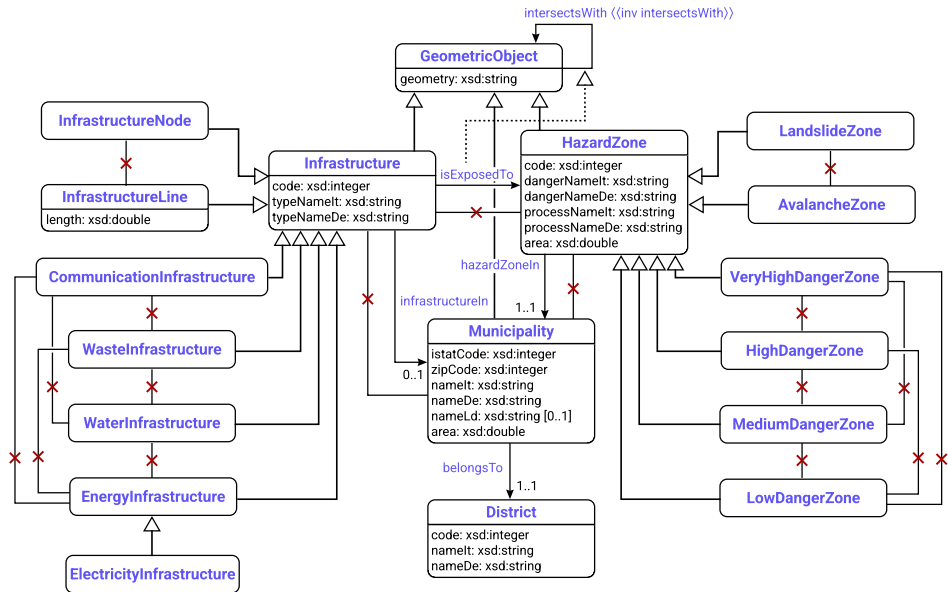
- ▶ **Key Object Properties:**

- `intersectsWith`: Symmetric topological overlap.
- `isExposedTo` $\sqsubseteq$ `intersectsWith`: **Links** `Infrastructure` **directly to** `HazardZone`.
- `infrastructureIn`, `hazardZoneIn`, `belongsTo`: **Administrative links**.

- ▶ **Data Properties:**

- Encapsulate WKT geometry strings, area and length attributes, localized names, and codes.

Domain Ontology Modeling – Cont.



Virtual Knowledge Graph Mappings

► IRI Design:

- Stable URI templates generated using table names and primary keys.
- e.g., :infrastructure-line/{code}, :municipality/{istat_code}.

MAP-DISTRICT

Source: SQL Query

```
SELECT code, name_it, name_de
FROM District
```

Target: Triples Template

```
:district/{code} a :District ;
:districtCode {code}^^xsd:integer ;
:districtNameIt {name_it}^^xsd:string ;
:districtNameDe {name_de}^^xsd:string .
```

MAP-MUNICIPALITY

Source: SQL Query

```
SELECT istat_code, zip_code, distr_code,
       name_it, name_de, name_ld,
       ST_Area(ST_MakeValid(geom)::geography) AS
         area
       ST_AsText(geom) AS geom
FROM Municipality
```

Target: Triples Template

```
:municipality/{istat_code} a :Municipality ;
:istatCode {istat_code}^^xsd:integer ;
:zipCode {zip_code}^^xsd:integer ;
:nameIt {name_it}^^xsd:string ;
:nameDe {name_de}^^xsd:string ;
:nameLd {name_ld}^^xsd:string ;
:area {area}^^xsd:double ;
:geometry {geom}^^xsd:string ;
:belongsTo :district/{distr_code} .
```

More Advanced VKG Mappings

- ▶ Normalized Relational Joins:
 - SQL joins to reconstruct localized data properties for normalized entities.
- ▶ Semantic Sub-classification:
 - SQL `WHERE` filter mappings to auto-classify instances into subclasses.
- ▶ Spatial Object Property Generation:
 - Utilized PostGIS spatial functions inside SQL mapping queries.
 - Calculated real-world length or area metrics on the fly.

Interactive Web Application

- ▶ Tech Stack:

- Single-Page Web Application built with React, Vite, React Router, and Leaflet Maps.

- ▶ SPARQL Integration:

- Communicates directly with the Ontop Virtual Knowledge Graph endpoint.

- ▶ Key Application Features:

- Instant answers to some preconfigured domain queries.

- Dynamic query generated based on custom filter configuration.

- Synchronized interactive map and paginated data table for results.

- Embedded code viewer displaying the dynamically constructed SPARQL query.

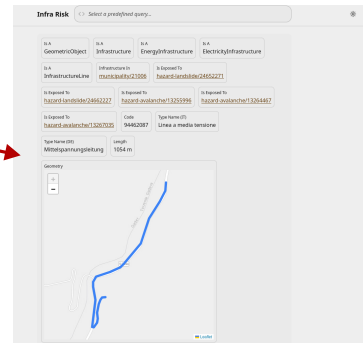
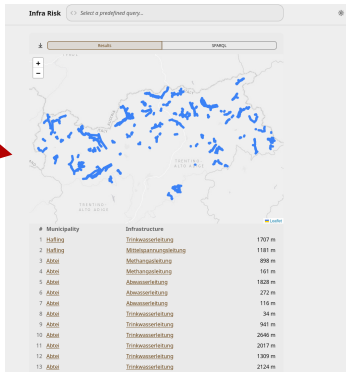
- Dedicated detail view for individual entities.

Live Demonstration

- ▶ Live Demo: <https://dc.rolandb.com>

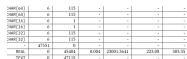
Infrastructure Risk Exposure Analyzer

- Select a predefined query...
- What are all of the municipalities?
 - What are all of the infrastructure assets?
 - What are all of the hazard zones?
 - Which infrastructure assets are exposed to hazards?
 - Which water-related infrastructure nodes are located w...
 - What are the infrastructure lines that intersect with aw...
 - In a given Municipality, how many electricity lines are c...
 - Which municipalities have the highest count of infrastr...

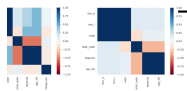


Conclusion

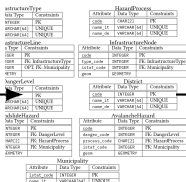
- ▶ Successfully implemented an end-to-end pipeline:
 - 1. Systematic profiling of 5 raw datasets using Metanome and YData.
 - 2. Redundancy removal and 3NF normalization in a PostGIS spatial database.
 - 3. Domain modeling with an OWL 2 QL ontology.
 - 4. On-the-fly topological reasoning using Ontop spatial VKG mappings.
 - 5. User-friendly React web interface for risk assessment.



i analysis results for the different columns of the `hr_infrastructure_line` table extracted
CTP since then.



616 `he_infrastructure_line` (B) Correlation matrix for `he_infrastructure_solid`



of Energy/Infrastructure. Note that while most of the infrastructure types can be directly associated with one of the four thematic sub-classes, some infrastructure elements may not belong to any of these thematic sub-classes.

[illegible]